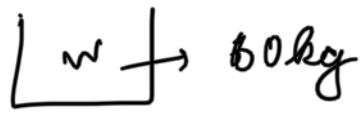
0/1 Knapsack Problem



40	10	50	90
10	20	40	30
0/1	0/1	0/1	0/1

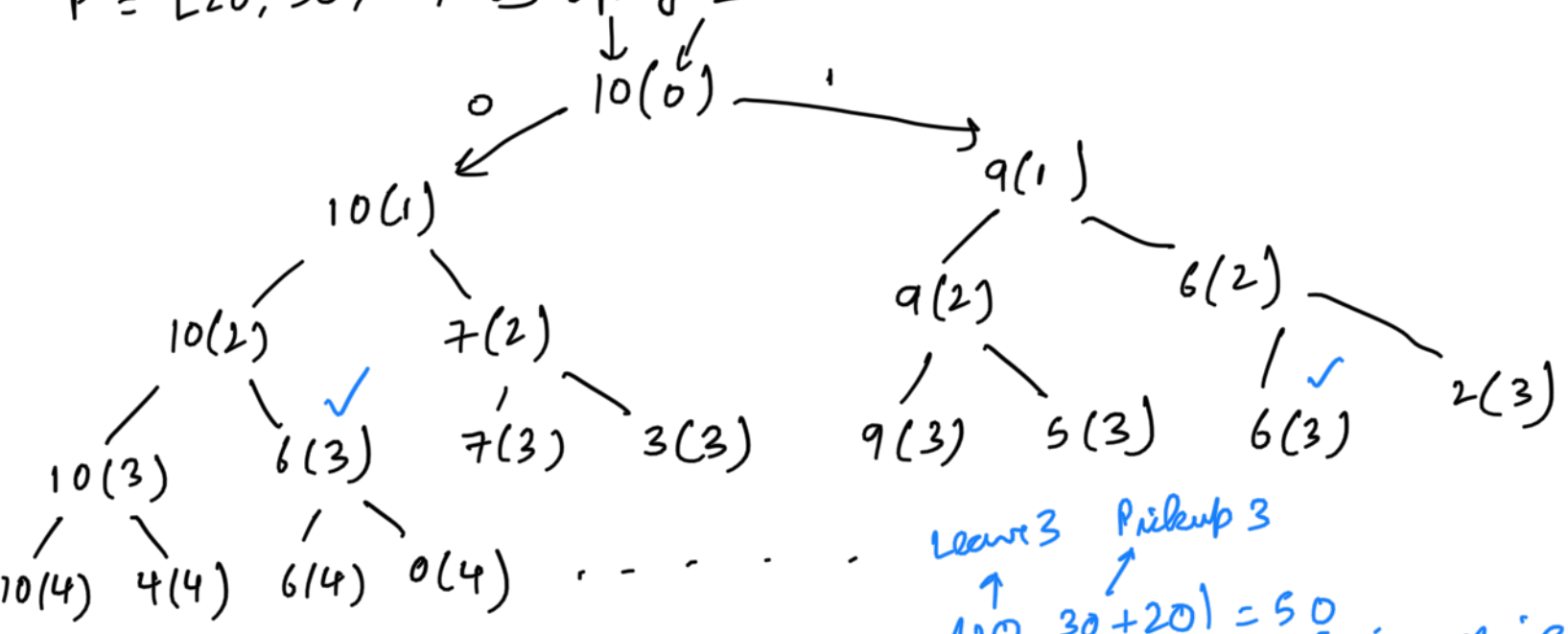
0 → leave
1 → Take

↑ P
0000
0001
⋮
→ 2^n

Profit[ ] = {60, 100, 120}

capacity = 50

W = 10
w = [1, 3, 4, 6] Weight[ ] = {10, 20, 30}
P = [20, 30, 10, 50] Capacity Index



Leave 3 Pickup 3

max(20, 30+20) = 50
↳ Remaining Weight

Capacity →

Weight	0	1	2	3	4	5	6	7	8	9	10
0 0	0	0	0	0	0	0	0	0	0	0	0
20 1	0	20	20	20	20	20	20	20	20	20	20
30 3	0	20	20	30	50	50	50	50	50	50	50
10 4	0	20	20	30	50	50	50	50	60	60	60
50 6	0	20	20	30	50	50	50	70	70	80	100

↑ Weight

max(50, 50+20)
Leave Take

max(60, 50+50)
= 100

Profit = 50, 30, 20
Weight = 6 3 1 = 10

Remaining capacity is memoized.
(Efficiency)

DYNAMIC PROGRAMMING

Dynamic programming applies when the sub-problems overlap.